

# Andrés F. Ramírez-Mejía

BSc, MSc, PhD

## Curriculum Vitae

December 2025

- 📍 Department of Ecology, Zoology and Genetics, Institute of Biology, Universidade Federal de Pelotas, Pelotas (RS), Brasil.
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## Education

- July 2008 – September 2014 **B.Sc. Biology**  
University of Caldas (Colombia)
- August 2015 – August 2017 **MSc. Conservation and Use of Biodiversity**  
Pontifical Javeriana University (Colombia)
- August 2018 – August 2023 **Ph.D. Biological Sciences**  
National University of Tucumán (Argentina)

## Current position

**May 2024 - present.** Postdoctoral researcher with double affiliation: Department of Ecology, Zoology and Genetics, Institute of Biology, Universidade Federal de Pelotas; Department of Zoology & Physiology, University of Wyoming. *Projects:* Army Ant Followers Project, and Hawaii-VINE Project.

## Past positions

- **Jul. 2023 - Apr. 2024.** Postdoctoral researcher at the Institute of Regional Ecology, UNT-CONICET. **Project:** Control systems for pollination service management in blueberry crops.

## Research interests

I am an ecologist with a deep curiosity for understanding how species interactions and functional diversity mediate ecological processes. My background has involved mammals, birds, and pollinators as model systems and the use of quantitative methods and field experiments to address my questions.

- Ecological interactions
- Pollination and frugivory
- Ecosystem services
- Community ecology
- Functional diversity
- R programming
- Bayesian statistics

## Technical skills

- Experimental design and data collection.
- Causal inference.
- Data analysis: frequentist and Bayesian statistics (lm, glm, glmm, hierarchical models), Gaussian process models, time series analysis, Bayesian networks, structural equation modeling, network analysis, and simulation studies.
- Programming languages: R (advanced), Python (basic) and Stan (basic).
- Spatial analysis tools: Google Earth Engine (basic), ArcGis (basic) and R packages: *terra*, *raster*, and *sf* (basic).
- Other tools: Git and GitHub for version control. RMarkdown and Quarto for report generation.

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## Publication, theses and oral presentations

### Published articles

- **Ramírez-Mejía A. F.**, Chacoff N., Cavigliasso P., Blendinger P. (2024). How much is enough? Optimizing beehive stocking densities to maximize the production of a pollinator-dependent crop. *Ecological modelling*, vol 498, article number: <https://doi.org/10.1016/j.ecolmodel.2024.110891>.
- Monasterologo, M, **Ramírez-Mejía A. F.**, et al. (2024). Animal pollination contributes to more than half of Citrus spp. production despite the species and cultivar. *Scientific reports*, vol 14, Article number: 22309. <https://www.nature.com/articles/s41598-024-73591-6>.

- Fontanarrosa G., Zarbá L., ... **Ramírez-Mejía A. F.** ... Piquer-Rodríguez M. (2024). Over twenty years of publications in Ecology: Over-contribution of Women reveals a new dimension of gender bias. *PLOS ONE*. <https://doi.org/10.1371/journal.pone.0307813>
- **Ramírez-Mejía A F**, Chacoff N, Lomáscolo S, Woodcock B, Schmucki R & Blendinger P. (2024). Optimal pollination thresholds to maximize blueberry production. *Agriculture Ecosystems and Environment* <https://doi.org/10.1016/j.agee.2024.108903>
- **Ramírez-Mejía A F**, Blendinger P, Woodcock B, Schmucki R, Escobar L, Morton R, Vieli L, Nunes-Silva P, Lomáscolo S, Morales C, Murúa M, Agostini K, & Chacoff N. (2023). Landscape structure and farming management interacts to modulate pollination supply and crop production in blueberries. *Journal of Applied Ecology*. DOI: <https://doi.org/10.1111/1365-2664.14553>
- Nunes-Silva Patrícia, **Ramírez-Mejía Andrés F.**, Blochtein B, Ramos J, Agostini K, Vieli L, Santanna M, Raguse-Quadros M, Maureen M., Chacoff N P, Cavigliasso P, Blendinger P G., Domingos S. (2023). Blueberry: pollination and production in South America. ISBN: 978-65-00-65347-2. DOI: <https://doi.org/10.5281/zenodo.7770381>.
- **Ramírez-Mejía A F**, Lomáscolo S & Blendinger P. (2023) Hummingbirds, honeybees, and wild insect pollinators affect yield and berry quality of blueberries depending on cultivar and farm's spatial context. *Agriculture Ecosystems and Environment*. DOI: <https://doi.org/10.1016/j.agee.2022.108229>
- Blendinger P G, Rojas T N, **Ramírez-Mejía A F**, Bender I M A, Lomáscolo S, Magro J, Núñez Montellano M G, Ruggera R A, Valoy M & Ordano M. (2022) Nutrient balance and energy-acquisition effectiveness: do birds adjust their fruit diet to achieve intake targets? *Functional Ecology*. DOI: [doi.org/10.1111/1365-2435.14164](https://doi.org/10.1111/1365-2435.14164)
- **Ramírez-Mejía A F**, Urbina-Cardona N, & Sánchez F. (2022). The interplay of spatial scale and landscape transformation moderates the abundance and intraspecific variation in the ecomorphological traits of a phyllostomid bat. *Journal of Tropical Ecology*. 38(1), 31-38. DOI: <https://doi.org/10.1017/S026646742100047X>
- **Ramírez-Mejía A F**, Echeverry-Galvis M A, & Sánchez F. (2021). Activity and habitat use by understory birds in a native Andean forest and a eucalypt plantation. *Wilson Journal of Ornithology*. 132(3): 721-729. DOI: [doi.org/10.1676/19-54](https://doi.org/10.1676/19-54)
- **Ramírez-Mejía A F**, Urbina-Cardona N, & Sánchez F. (2020) Functional diversity of phyllostomid bats in an urban-rural landscape: a scale-dependent analysis. *Biotropica*. 52(6): 1168-1182. DOI: <https://doi.org/10.1111/btp.12816>.
- **Ramírez-Mejía A F**, & Sánchez F. (2016). Activity patterns and habitat use of mammals in an Andean forest and a Eucalyptus reforestation in Colombia. *Hystrix, the Italian Journal of Mammalogy*, 27(2): 104-110. DOI: <https://doi.org/10.4404/hystrix-27.2-11319>.
- **Ramírez-Mejía A F**, & Sánchez F. (2015). Non-volant mammals in a protected area on the Central Andes of Colombia: new records for the Caldas department and the Chinchiná River basin. *Check List*, 11(2):1-6, Article 1582. DOI: <https://dx.doi.org/10.15560/11.2.1582>.

#### Accepted

- Juliana Wolter ... **Ramírez-Mejía A. F.**..., Rafael Dias. Intrinsic and extrinsic drivers shape frugivore functional diversity within a tree population. *Journal: Ecosphere*. **Role: coauthor, data analysis**

#### Under review

- **Ramírez-Mejía A. F.**, Mary De Aquino, Michael Castaño Díaz, Juliana Hinz Wolter, Henry S. Pollock, J. Patrick Kelley, Jeferson Vizentin-Bugoni & Corey E. Tarwater. Biotic and abiotic factors directly and indirectly impact the structure of non-trophic networks across a rainfall gradient. *Journal: Ecology*. **Role: first author**
- Laura Gómez-Murillo, Jeferson Vizentin-Bugoni, **Ramírez-Mejía A. F.**, Corey E. Tarwater. Rainfall alters network structure while fragmentation results in the breakdown of a mixed-species group of birds. *Journal: Oecologia*. **Role: coauthor, data analysis**
- Elizabeth J. Howard, **Andrés F. Ramírez-Mejía**, Michael Castaño, Mary De Aquino, Henry S. Pollock, Jefferson Vizentin-Bugoni, and Corey E. Tarwater. Testing long standing assumptions about army ant swarms: how profitability and dominance alter space use in birds. *Journal: Behavioral Ecology*. **Role: coauthor, data analysis**
- Jordan, Kimberley; **Ramírez-Mejía, Andrés F.**; Wilcox, Rebecca C.; Brawn, Jeffrey ; Alfonso, Camilo; Tarwater, Corey. Testing the space-for-time assumption and the hygric niche hypothesis using precipitation effects on avian body condition. *Journal: Proceedings of the Royal Society B: Biological Sciences*. **Role: coauthor, data analysis**
- **Andrés F. Ramírez-Mejía**, Corey E. Tarwater, J. Patrick Kelley, Jinelle H. Sperry, Jeffrey T. Foster, Donald R. Drake, and Jeferson Vizentin-Bugoni. Temporal dynamics of seed-dispersal networks: Disentangling the role of direct biotic and abiotic factors and bottom-up processes. *Journal: Ecology*. **Role: first author**
- **Andrés F. Ramírez-Mejía**, Francisco Sánchez, J. Nicolás Urbina-Cardona & Natalia Ladino. Functional divergence drives the prevalence of low-abundance species in bat assemblages. *Journal: Functional Ecology*. **Role: first author**

#### Articles in preparation

- Mauro Galetti,... **Ramírez-Mejía A. F.**... Measuring the macroecological and local effects on plant-animal interaction using artificial fruits. **Role: coauthor, data analysis**

- **Ramírez-Mejía A. F.**, A. Carolina Monmany, Roxana Aragón, Carolina Cuezco, Natalia Ladino, Natalí Villalba, A. Carolina Jozami, Alberto Slanis, Mariano Lucia, Leopoldo Alvarez & Natacha P. Chacoff. Low opportunity cost areas in large-scale monoculture for pollinator conservatio. **Role: first author**
- Mary De Aquino<sup>1</sup>, Henry S. Pollock, J. Patrick Kelley, **Andrés F. Ramírez-Mejía**, Michael Castaño Díaz<sup>1</sup>, Jeferson Vizontin-Bugoni & Corey E. Tarwater. Species' roles, specialization, and the factors that influence roles in mixed-species animal groups. **Role: review**

#### Code and online datasets

- **Ramírez-Mejía A. F.**, Chacoff N., Cavigliasso P., & Blendinger P. (2024). How much is enough? Optimizing beehive stocking densities to maximize the production of a pollinator-dependent crop. *Ecological modelling*, vol 498, article number: <https://doi.org/10.1016/j.ecolmodel.2024.110891>. GitHub repository: [https://github.com/andres-frm/postdoc\\_cropPol](https://github.com/andres-frm/postdoc_cropPol)
- **Ramírez-Mejía, Andrés F.** (2023). Landscape structure and farming management interacts to modulate pollination supply and crop production in blueberries [code]. **Journal of Applied Ecology**. Zenodo. <https://doi.org/10.5281/zenodo.10070657>
- **Ramírez-Mejía, Andrés F.** et al. (2023). Landscape structure and farming management interacts to modulate pollination supply and crop production in blueberries [Dataset]. **Journal of Applied Ecology**. Dryad. <https://doi.org/10.5061/dryad.bg79cnp>
- **Ramírez-Mejía, A F.**; Urbina-Cardona, J N; Sánchez, F (2020). Data from: Functional diversity of phyllostomid bats in an urban-rural landscape: a scale-dependent analysis [Dataset]. **Biotropica**. Dryad. <https://doi.org/10.5061/dryad.sn02v6x1p>

#### Theses

**PhD.** 2018 - 2023. The step-by-step of pollination service in blueberry crops: patterns and mechanisms. Thesis advisors: Pedro G. Blendinger, PhD; independent research CONICET, assistant professor UNT. Silvia Lomáscolo, PhD; assistant researcher CONICET.

**MSc.** 2015 - 2017. Seed dispersion and functional diversity of phyllostomid bats in an anthropogenic landscape: a multi-scale approach. Thesis advisors: Francisco Sánchez, PhD; Universidad de los Llanos, Colombia. J. Nicolás Urbina-Cardona, PhD; Pontificia Universidad Javeriana, Colombia.

**BSc.** 2008 - 2014. Activity patterns, habitat use, and mammals richness in native and reforested forests at the Colombian Andes. Thesis advisor: Francisco Sánchez, PhD; Universidad Militar Nueva Granada, Cundinamarca, Colombia.

#### Teaching experience

|      |                                                      |                                            |
|------|------------------------------------------------------|--------------------------------------------|
| 2024 | Introduction to R programming (Level: graduate)      | Universidade Federal de Pelotas (Brazil)   |
| 2021 | Data Clinic (Level: graduate)                        | National University of Tucumán (Argentina) |
| 2019 | Introduction to R programming (Level: undergraduate) | National University of Tucumán (Argentina) |

#### Graduate courses

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|------|----------------------------------------------------------|---------------------------------------------|-----|-----------|
| 2023 | Google Earth Engine - Integral                           | REDES Center (CONICET)                      | 60h | Argentina |
| 2022 | Introduction to Bayesian statistics methods in ecology   | University of Buenos Aires                  | 45h | Argentina |
| 2022 | Ecosystem services: concepts, tools and perspectives     | National University of Cuyo                 | 45h | Argentina |
| 2022 | Data science with Python                                 | National University of Southern Patagonia   | 40h | Argentina |
| 2021 | Models and data in ecology                               | National University of Comahue              | 40h | Argentina |
| 2021 | Statistical tools for addressing gender metrics          | National University of Tucumán              | 40h | Argentina |
| 2020 | Ecological networks                                      | University of São Paulo                     | 60h | Brazil    |
| 2019 | Pollination Ecology                                      | National University of Comahue              | 60h | Argentina |
| 2019 | 13° International Pollination Course                     | Federal University of Jequitinhonha Valleys | 98h | Brazil    |
| 2019 | Advanced statistical methods in ecology and evolution    | University of the Uruguay Republic          | 60h | Uruguay   |
| 2018 | Advanced statistical models using programming language R | National University of Córdoba              | 40h | Argentina |

#### Fellowships

- **2023.** Postdoctoral fellowship financed by CONICET, Argentina. RESOL-2023-1026-APN-DIR#CONICET.
- **2017.** PhD fellowship financed by CONICET, Argentina. N° 4122. Thesis: Plant-animal interaction in blueberry crops (*Vaccinium corymbosum*, Ericaceae) and indirect effect with the Yungas forest.

## **Distinctions**

- 2017 Academic merit award. During MSc degree at Pontifical Javeriana University
- 2017 Graduated with honors CUM LAUDE. During MSc degree at Pontifical Javeriana University
- 2017 MSc thesis with honorary mention. During MSc degree at Pontifical Javeriana University

## **Languages**

- Spanish
- English
- Portuguese